DIC11
Interconnect RC Delay and Capacitive Coupling

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives and specification mapping

Lab 11 (PDF pages 44–51) requests wire-RC analysis, driver sizing, wire-length sweeps and adjacent-wire coupling comparisons. This implementation retains the handout's three wire-parameter sets and corresponding device W/L dimensions: 45 driver-size cases, 30 length cases and 180 coupling cases, totaling three netlists and 255 completed cases.

The handout labels these sets as 0.35/0.25/0.18 μm processes at 3.3/2.5/1.8 V. The corresponding three MOS process models are unavailable. Here A/B/C identify RC and geometry scenarios; every scenario uses the same supplied model at 1 V. The results compare wire and sizing effects, and do not reproduce absolute delays for three foundry processes.

2 Circuit, parameters and hand calculations

The handout's three-section π model uses three resistors $R/3$, endpoint capacitors $C/6$ and two interior capacitors $C/3$, preserving total resistance and capacitance. Inv1 drives the wire and Inv2 is the receiver. Delays t_1 and t_2 reach the driver output and receiver input with inversion; t_3 reaches the receiver output with the same polarity as the original input.

The 10 mm wire values from Table 11.1 and recalculated RC products are below. Set A's printed 3.17 ns is inconsistent with $1.7 \times 2.11 = 3.587$ ns. The simulation retains the R and C columns and uses their actual product; the printed value is not treated as a measurement.

Set	R (kΩ)	C (pF)	RC (ns)	$W_p/W_n/L$ (μm)
A	1.70	2.11	3.587	2.7 / 0.9 / 0.35
B	2.38	2.39	5.688	1.8 / 0.6 / 0.25
C	3.39	2.46	8.339	1.5 / 0.5 / 0.18

Driver widths scale by N=1–15 while receiver widths stay at one unit. The length sweep fixes N=5 and covers 1–10 mm in 1 mm increments. Both wire R and C scale with length, so the internal wire-RC contribution grows approximately quadratically.

The coupling model contains left, center and right wires, with the center as victim. Every section retains both ground and interwire capacitances. Table 11.4 parameters per micrometer are:

Set	R (Ω/μm)	C_c (fF/μm)	C_g (fF/μm)	C_s (fF/μm)
A	.17	.08	.07	.14
B	.24	.09	.05	.14
C	.34	.10	.04	.14

For each scenario, victim rise/fall, six neighbor patterns R/R, R/static, R/F, static/static, static/F and F/F, and lengths 2/4/6/8/10 mm form the full $3\times2\times6\times5 = 180$-case matrix. Static neighboring inputs remain at 0 V. R/F describe transitions at the original input of each wire's driver.

3 Simulation setup and measurement definitions

The separately supplied MOS model is evaluated at 1.0 V and 27 °C, retaining the handout topology and the comparison conditions below. One volt is a simulation setting, not a device rating established by available model documentation. Numerical results from the handout's older processes are not acceptance targets for this model.

HSPICE S-2021.09 runs with RUNLVL=6. Tables use actual native measurements and figures use actual printed numerical traces. Every variant resets all swept parameters and temperature, avoiding inherited settings from an earlier variant. Completed solutions and measurements do not establish process reliability. Internal warnings in encrypted model blocks remain incompletely visible. Uncoupled tests use 100 ps input ramps, a 180 ns transient and 20 ps printing, including both polarities. Coupling tests apply a single 100 ps ramp at 20 ns and run to 120 ns. Delay is the difference between original-input and receiver-output 50% crossings. The long observation window accommodates the slowest wire transitions.

4 Driver-size and length results

Larger drivers improve near-end delay, while far-end delay increasingly reaches the limit imposed by the wire itself.

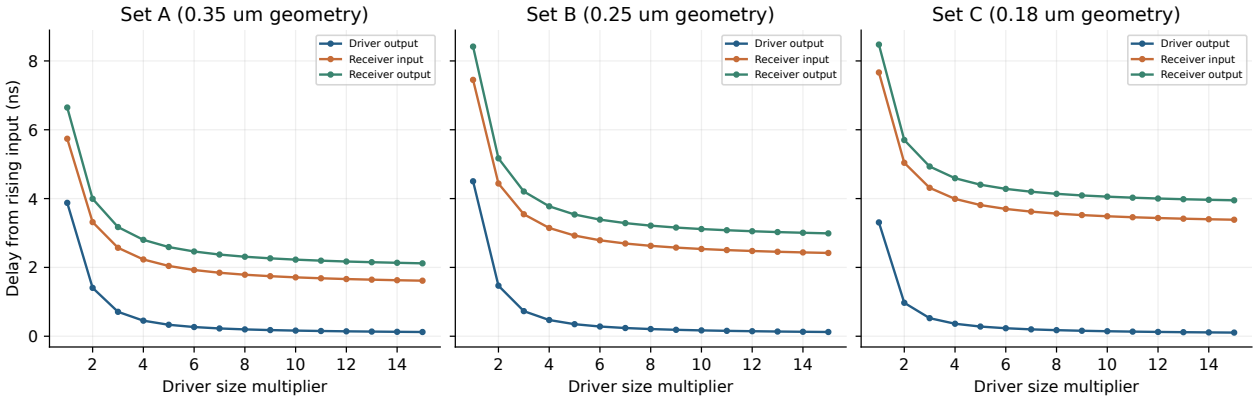


Figure 1: Three observation points on 10 mm wires; curves use the rising original input.

Set	N=1, t_3 (ns)	N=5, t_3 (ns)	N=15, t_3 (ns)
A	6.646	2.591	2.118
B	8.417	3.537	2.989
C	8.476	4.403	3.950

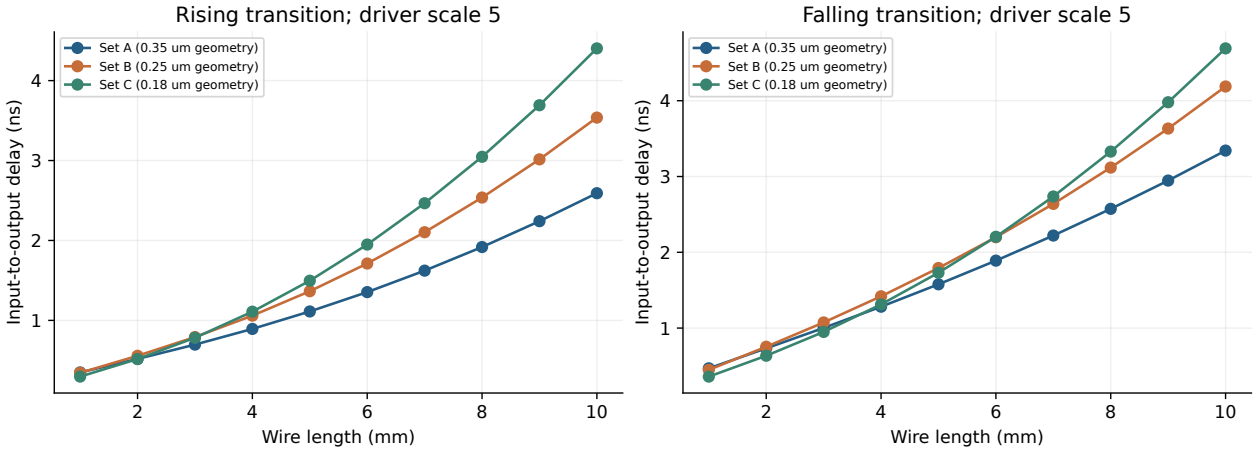


Figure 2: Receiver-output delay versus wire length at N=5 for both polarities.

5 Coupling results

The table compares the 10 mm extremes of both neighbors switching with the victim or against it. Every value is from the current simulations.

Set	Victim	Same (ns)	Opposite (ns)	Ratio
A	R	1.186	3.978	3.35
A	F	1.576	6.064	3.85
B	R	1.278	5.477	4.29
B	F	1.549	6.990	4.51
C	R	1.464	7.424	5.07
C	F	1.588	8.069	5.08

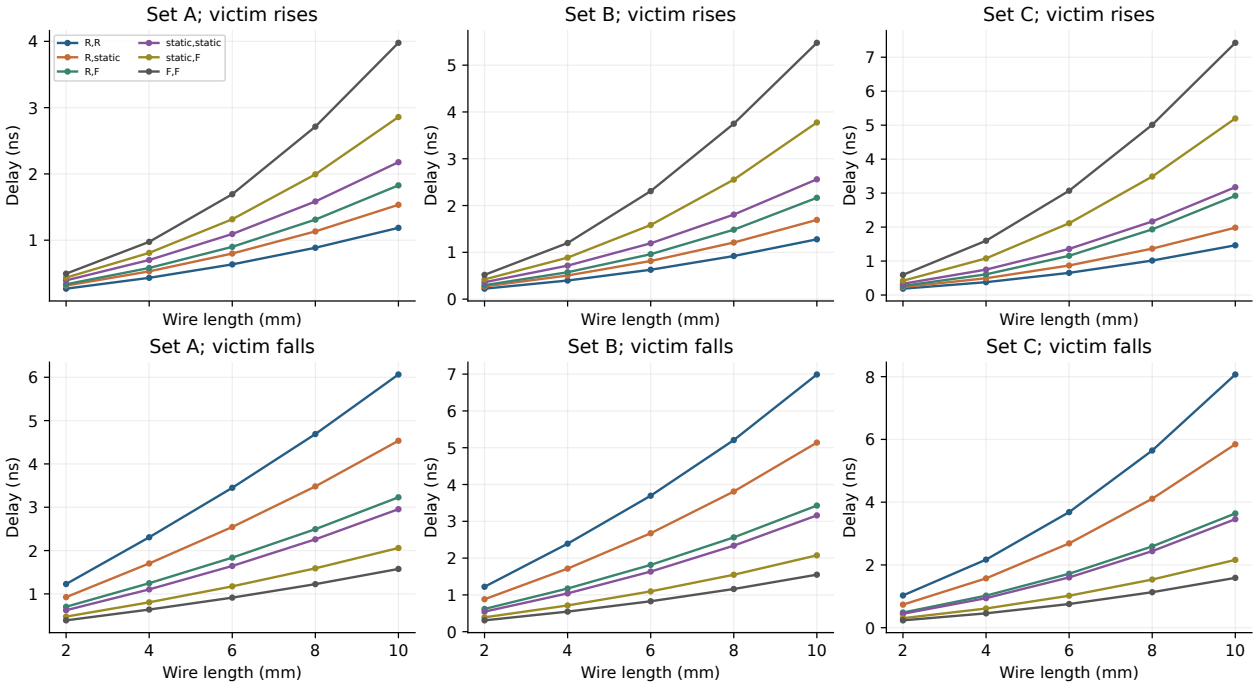


Figure 3: All six neighbor patterns; R/F/static are defined at the driver inputs.

6 Discussion and questions

When adjacent wires move together, the interwire capacitor sees a smaller differential change; opposite transitions increase that change and the effective load. For ideal synchronized ramps, $C_{eff} = C_g + \sum C_c(1 - dV_{adj}/dV_{victim})$, so each neighbor contributes approximately 0, C_c or $2C_c$ for same-direction, static or opposite-direction motion. Real slopes and timing differ, so delay ratios need not equal these capacitance ratios.

Lower-resistivity metal, thicker or wider wires, lower-permittivity dielectrics, increased spacing and shorter routes can reduce wire delay, with tradeoffs in area, capacitance and routing resources. Circuit options include appropriately sized repeaters and tapered drivers. Oversizing only the first driver loads its predecessor and cannot remove the wire's intrinsic RC. Shielding reduces crosstalk but adds ground capacitance and consumes metal resources.

All RC and coupling scan conditions are complete. Absolute comparisons across actual fabrication processes remain limited by missing matching MOS models. Discontinuous early numbers in the handout Table 11.5(c) are not adopted as validation targets.

7 Reproducibility files

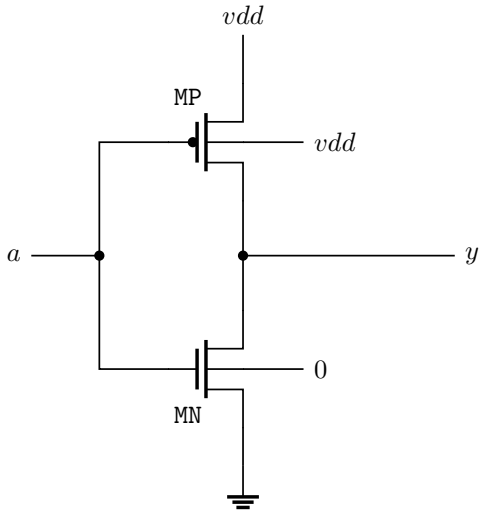
The following netlists accompany this report. A comment marks the local model setup; neither the model file nor its path is published. A valid local model configuration must be supplied before execution.

- `01_driver_sizing.sp`
- `02_wire_length.sp`
- `03_coupled_wires.sp`

The lab results directory contains every case measurement, representative waveform CSV files, derived metrics and a specification manifest. Waveform exports retain at most 2501 uniformly selected samples; direct measurements are not downsampled. Complete original listings remain local.

8 MOS transistor schematic

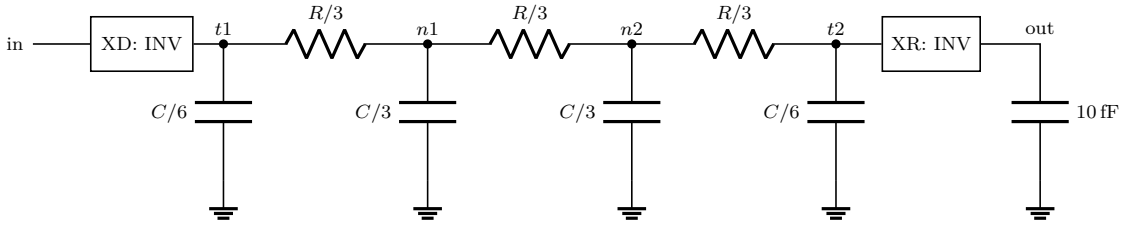
8.1 Driver and receiver transistor cell



All three public decks use INV: PMOS S/B connect to vp=vdd, NMOS S/B to vn=0. Every INV block below expands to these two MOS devices.

Drivers use $W_n = WNBASE \cdot SIZE$ and $W_p = WPBASE \cdot SIZE$; receivers use SIZE=1; $L = LBASE$. Sets A/B/C use $(WNBASE, WPBASE, LBASE)$ equal to $(0.9, 2.7, 0.35)$, $(0.6, 1.8, 0.25)$ and $(0.5, 1.5, 0.18)\,\mu$m. These are size/RC data sets using one supplied model, not three process models.

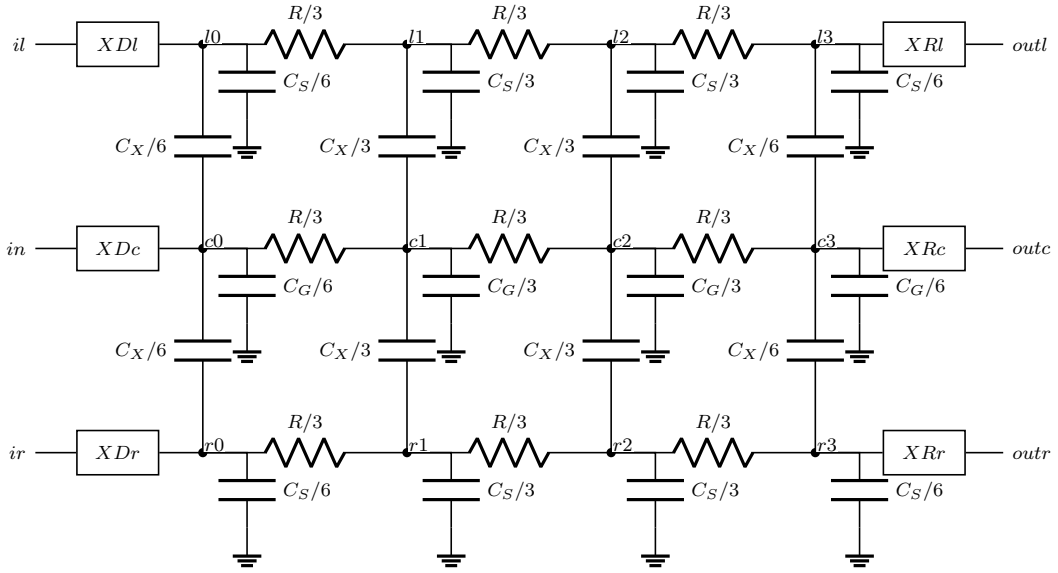
8.2 Three-section pi network for one wire



Actual connectivity in `01_driver_sizing.sp` and `02_wire_length.sp`: $R = RUNIT \cdot LEN$, $C = CUNIT \cdot LEN \cdot 1$ fF.

The supply is 1 V. The first deck fixes LEN=10000 μm and sweeps SIZE=1–15; the second fixes SIZE=5 and sweeps LEN=1000–10000 μm. The receiver output has an additional 10 fF; all four wire nodes have shunt capacitance.

8.3 All nodes of the three coupled wires



`03_coupled_wires.sp`: l/c/r denote left/victim/right wires. Each wire has three resistors and four ground capacitors. Adjacent wires have a coupling capacitor at every matching node index.

$R = RUNIT \cdot LEN$; $C_S = CSUNIT \cdot LEN$ fF, $C_G = CGUNIT \cdot LEN$ fF, $C_X = CCUNIT \cdot LEN$ fF. All XD drivers use SIZE=5 and all XR receivers SIZE=1, with the size sets above. This deck adds no CL at outl/outc/outr; the previous circuit's 10 fF is not part of this topology.

Inputs in, il and ir are three ground-referenced behavioral voltage sources. The victim uses $(1 - CEN)/2 + CEN \cdot V(base)$; neighbor coefficients 1 and -1 select rising/falling transitions, while coefficient 2 holds the input at 0 V. The base signal rises from 0 to 1 V during 20–20.1 ns. Only l–c and r–c coupling capacitors exist; there is no direct l–r connection.